Nihanth Baratam

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Professional Summary

Full-Stack Software Engineer at Solytics Partners building **MRM Vault**, a model risk management platform used by financial institutions. I ship features end to end: **Django** APIs, **PostgreSQL** data models and migrations, **Redis** caching on the backend, and **React 18 / Redux Toolkit** interfaces on the frontend. I focus on performance and reliability, including faster API responses through query optimisation and caching, fixes for authentication and authorization issues, and safe schema changes on production data. B.Tech in Computer Science from VIT (8.86 CGPA).

Professional Experience

Solytics Partners Pune, India
Associate Software Engineer (Full-Stack) | MRM Vault *Aug 2026 – Present*

- **End-to-End Features:** Own full-stack features for MRM Vault from requirement to release, designing **Django** REST APIs, **PostgreSQL** data models and the matching **React** UI.
- **API Performance:** Improved response times on high-traffic endpoints by ∼40% by removing N+1 queries with Django ORM `select_related`/`prefetch_related`, adding PostgreSQL indexes and caching hot reads in **Redis**.
- **Database & Migrations:** Wrote and shipped Django schema and data migrations against production PostgreSQL, planning backward-compatible rollouts so releases went out without downtime or data loss.
- **Auth & Security:** Diagnosed and fixed authentication and authorization issues across the API and UI, including session and token handling and role-based permission checks on protected endpoints.

Software Development Engineer Intern | MRM Vault *Feb 2026 – Jul 2026*

- **Frontend Architecture:** Delivered a production Risk Analysis UI with **React 18**, **Vite** and **MUI 5**, with **ECharts**, **Plotly** and **React Flow** visualising risk metrics and model dependency graphs, so stakeholders review model outputs 3× faster.
- **Performance:** Cut ∼40% of redundant API calls by introducing Redux Toolkit global state and TanStack Query caching, improving data consistency across sessions.
- **Reliability & Security:** Resolved 20+ critical bugs in the MRM platform; enforced form integrity with Formik/Yup and blocked XSS through DOMPurify sanitisation on all user-input surfaces.

Genpact (ThinkFCM) Hyderabad, India
Software Engineering Intern | Quality Engineering & Backend Validation *May 2025 – Jun 2025*

- **API Reliability:** Improved endpoint reliability across 3+ system integrations by building automated **Node.js** validation scripts that caught edge-case failures before each production deployment.
- **Defect Reduction:** Reduced defect resolution time by 30% across enterprise client modules through TDD, structured unit-test scenarios and root cause analysis tracked in Jira.

Technical Skills

Languages: Python, JavaScript (ES6+), SQL, Java, C/C++, Solidity
Backend: Django (REST APIs, ORM, Migrations), Node.js, Express.js, RESTful APIs, JWT/Auth, RBAC, Redis, Socket.io
Frontend: React 18, Redux Toolkit, TanStack Query, MUI 5, Vite, React Flow, ECharts, Plotly, Formik/Yup
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Redis, AWS S3, Docker, Kubernetes
Tools & Practices: Git, Postman, Jira, Agile/Scrum, Code Reviews, Unit Testing, DOMPurify (XSS)

Key Projects

Event Whiz | *Node.js, Express, MongoDB, Redis, JWT, AWS S3, React.js*

- **Latency:** Cut API response time from 800ms to 300ms (62%) with **Redis** caching on high-traffic endpoints, at no extra infrastructure cost.
- **Auth & Media:** Secured every API route with JWT authentication and Role-Based Access Control; reduced image-load overhead by 40% with AWS S3 pre-signed URLs.

VIT Campus Parking: Real-Time Reservation System | *Node.js, MongoDB, Socket.io, React.js*

- **Concurrency:** Eliminated double-booking across 50+ live slots in 5+ zones with atomic database transactions, holding 99% booking accuracy during peak-hour spikes.
- **Real-Time Sync:** Pushed live availability updates in under 200ms over Socket.io, cutting average booking time by 70% with no manual refreshes.

Education

Vellore Institute of Technology (VIT) 2022 – 2026
B.Tech in Computer Science (Blockchain) | CGPA: 8.86 / 10 *Vellore, India*

- Relevant coursework: Data Structures & Algorithms, DBMS (SQL), OOP (C++), Software Engineering

Sri Chaitanya Junior College 2020 – 2022
12th BIEAP | 94%

Achievements & Certifications

MERN Stack Certification (Ethnus): Completed full-stack training in MongoDB, Express.js, React and Node.js.
VIT Hackathon, 3rd Place: Designed and built a campus bus shuttle website that solved student transport tracking and scheduling.
Leadership: Club Lead, Chess & Board Games Club, VIT; organised and ran 20+ competitions for students.
DSA: Solved 300+ LeetCode problems across arrays, linked lists, trees and dynamic programming.